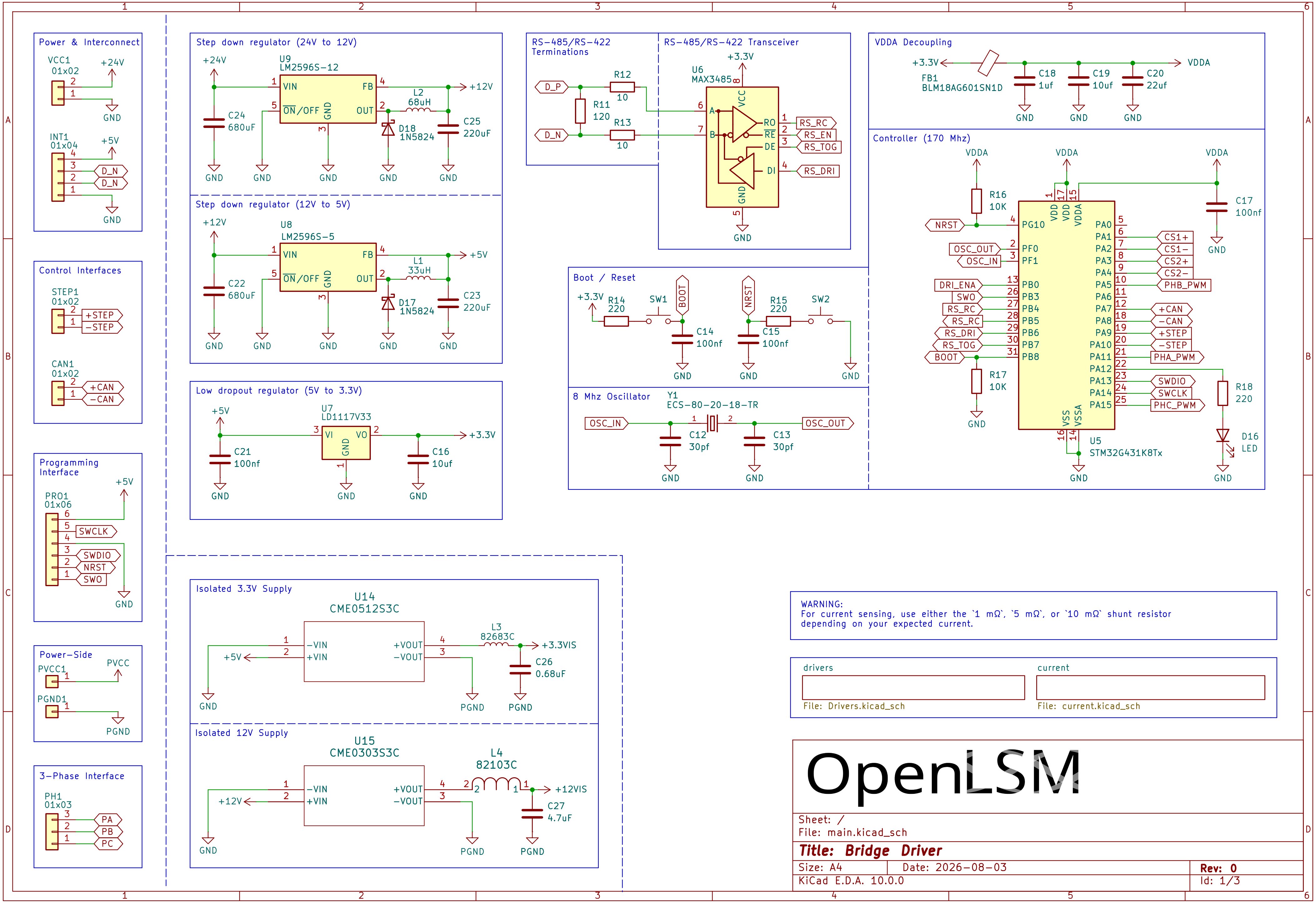
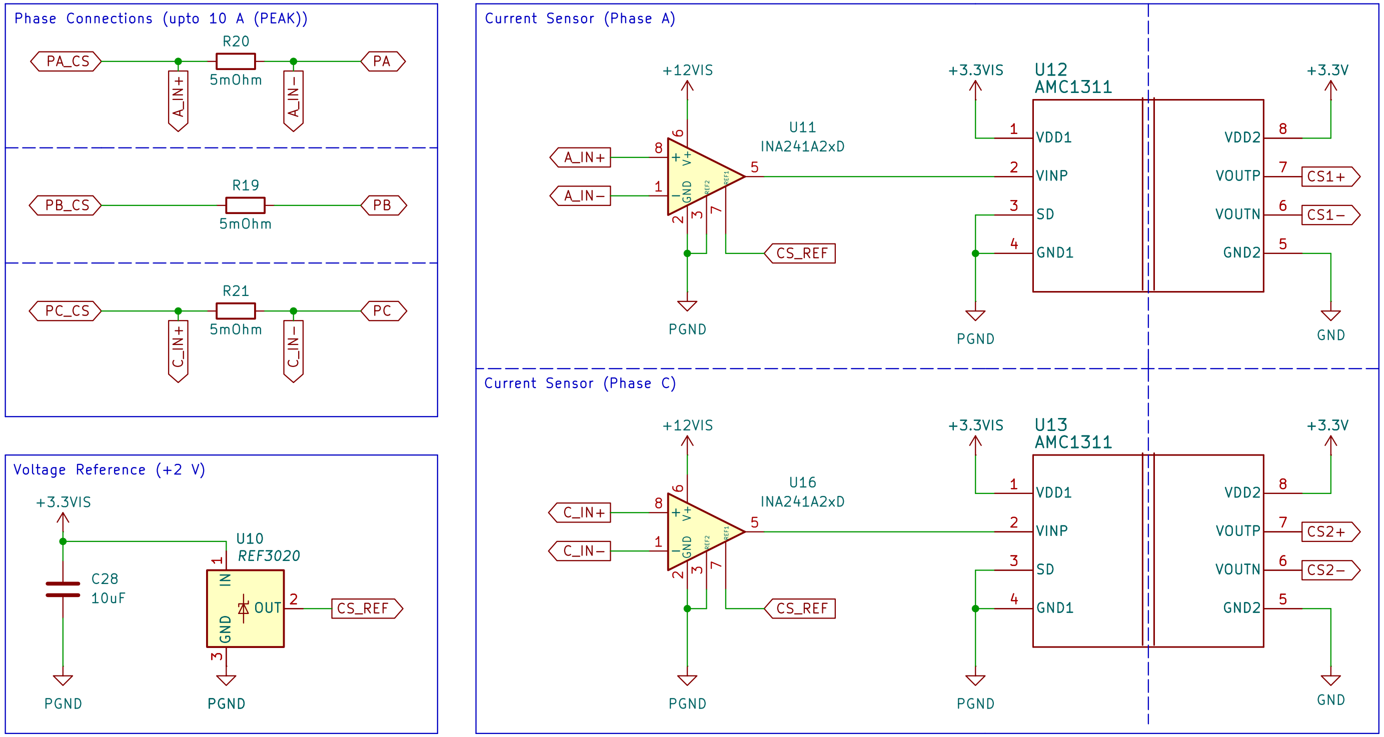

Power & Interconnect
VCC1 01x02 +24V GND
INT1 01x04 +5V D_N D_N GND
Control Interfaces
STEP1 01x02 +STEP -STEP
CAN1 01x02 +CAN -CAN
Programming Interface
+5V PRO1 01x06 SWCLK SWDIO NRST SWO GND
Power-Side
PVCC1 PVCC PGND1 PGND
3-Phase Interface
PH1 01x03 PA PB PC
Step down regulator (24V to 12V)
+24V U9 LM2596S-12 VIN FB ON/OFF GND OUT L2 68uH +12V C24 680uF D18 1N5824 C25 220uF GND
Step down regulator (12V to 5V)
+12V U8 LM2596S-5 VIN FB ON/OFF GND OUT L1 33uH +5V C22 680uF D17 1N5824 C23 220uF GND
Low dropout regulator (5V to 3.3V)
+5V U7 LD1117V33 VI VO GND +3.3V C21 100nf C16 10uf GND
RS-485/RS-422 Terminations
D_P D_N R12 10 R11 120 R13 10
RS-485/RS-422 Transceiver
+3.3V U6 MAX3485 VCC A B RO RE DE DI GND RS_RC RS_EN RS_TOG RS_DRI
VDDA Decoupling
+3.3V FB1 BLM18AG601SN1D C18 1uf C19 10uf C20 22uf VDDA GND
Controller (170 Mhz)
VDDA R16 10K NRST OSC_OUT OSC_IN DRI_ENA SWO RS_RC RS_RC RS_DRI RS_TOG BOOT R17 10K GND
PG10 PF0 PF1 PB0 PB3 PB4 PB5 PB6 PB7 PB8 VDD VDD VDDA VSS VSSA
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
CS1+ CS1- CS2+ CS2- PHB_PWM +CAN -CAN +STEP -STEP PHA_PWM SWDIO SWCLK PHC_PWM
U5 STM32G431K8Tx C17 100nf R18 220 D16 LED GND
Boot / Reset
+3.3V R14 220 SW1 BOOT C14 100nf NRST R15 220 SW2 C15 100nf GND
8 Mhz Oscillator
Y1 ECS-80-20-18-TR OSC_IN C12 30pf C13 30pf OSC_OUT GND
Isolated 3.3V Supply
U14 CME0512S3C -VIN +VIN +VOUT -VOUT L3 82683C +3.3VIS +5V C26 0.68uF GND PGND
Isolated 12V Supply
U15 CME0303S3C -VIN +VIN +VOUT -VOUT L4 82103C +12VIS +12V C27 4.7uF GND PGND
WARNING:
For current sensing, use either the `1 mΩ`, `5 mΩ`, or `10 mΩ` shunt resistor depending on your expected current.
drivers File: Drivers.kicad_sch
current File: current.kicad_sch
OpenLSM
Sheet: /
File: main.kicad_sch
Title: Bridge Driver
Size: A4 Date: 2026-08-03 Rev: 0
KiCad E.D.A. 10.0.0 Id: 1/3

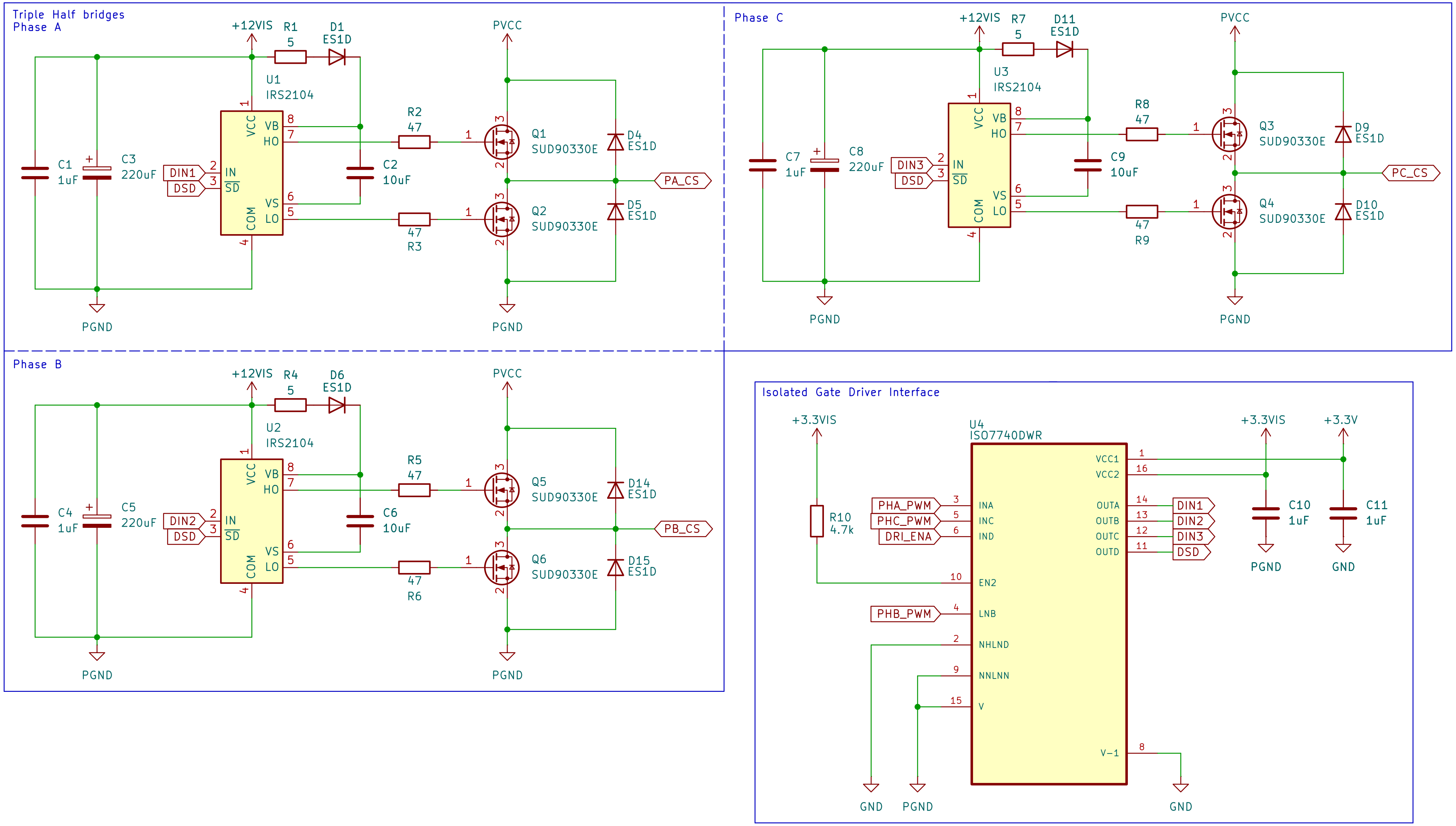


OpenLSM

Sheet: /current/
File: current.kicad_sch

Title: Current Sensors & Isolation

Size: A4	Date:	Rev: 0
KiCad E.D.A. 10.0.0		Id: 2/3



OpenLSM

Sheet: /drivers/
File: Drivers.kicad_sch

Title: Phase Drivers & Isolation

Size: A4 | Date: | Rev: 0
KiCad E.D.A. 10.0.0 | Id: 3/3